

Element A student work

Thursday, September 25, 2025 12:56 PM

Daniel Pollitt

Due: 4/2 @ 11:59pm

Presentation and Justification of the Problem

Universities often keep animals in their care to do scientific research on them. This research has helped protect hundreds of animal species, and allows us to learn how they work, so we can improve our own technology. However, a key aspect of these universities is their aim to make the animals as comfortable as possible.

Towson University is no exception, with a wide variety of animals under their care, from frogs, to spiders, to even axolotls. Their goal for this, like other universities, is to have these animals live in the best conditions possible and to the best of their lives. Two of these animals in captivity are a pair of turtles: a female Red-Eared Slider turtle, and a male Western Painted turtle.

While they are both turtles, they have very different behaviors and characteristics. However, one thing that they do have in common is that they both need to bask daily. The problem is, the Red-Eared Slider is often very pushy, monopolizing the basking spot. Not only that, but when the turtles are being fed, it will often aggressively try to steal food from its smaller cousin. All in all, a better basking platform onto which they can both fit, as well as a spot where the small Western Painted turtle can shelter from the Red-Eared Slider, are needed.

According to Lambert et al. (2019) and Pham (2025), the Red-Eared Slider is listed as one of the world's most invasive species. Pham went into further detail, stating that, starting off, they aggressively compete with the native turtle species for the best basking spot. In addition, Pham stated that Red-Eared Sliders outcompete the native species for food, which is why their invasiveness is such a problem. The current Red-Eared Slider turtle in captivity at Towson is named Bella, and measures 11 inches long and 6 inches wide. She currently shares a tank with "Zark", a Western Painted turtle who is much smaller, being only 6 inches long and 4 inches wide. According to stakeholders, including the Living Lab Scientist, Bella constantly tries to eat Zark's food and can sometimes be very pushy. In addition, Ray (2026) says that Western Painted turtles, when compared to Red-Eared Sliders, are much more shy. At the moment they believe that when they detect danger, they will slide into the water when basking. Ray also states that these turtles interact with each other in the wild, with the Red-Eared Slider displacing the Western Painted turtle. Finally, Gerkensmeyer (2025) states that while these

turtles like to live with other turtles, it is best that they have a place in which they can hide from the other turtles whenever they feel like it.

All in all the problem is that the Red-Eared Slider is acting in captivity like it does in the wild: invasively. In addition, since the Western turtle is already very skittish on its own, it would be beneficial for it to have its own little spot to get away from the Red-Eared Slider. This is not only important for the Western Painted turtle's health, as high levels of stress --- which prolonged unwanted contact with the Red-Eared Slider causes --- can be detrimental to the turtle's health, but also for the scientist in the Living Lab who is studying these turtles.

References

facebook. (2025, October 9). *Red-eared Slider: Characteristics, Habitat, Behaviors & Lifecycle*.

ANIMAL PEDIA. <https://animal-pedia.org/testudines/red-eared-slider/>

Gerkenmeyer, R. (2025, March 26). *Western Painted Turtle: Pictures, Facts, Diet & Care Guide | Hepper Pet Resources*. Hepper Pet Resources. <https://articles.hepper.com/western-painted-turtle/>

Lambert, M. R., McKenzie, J. M., Screen, R. M., Clause, A. G., Johnson, B. B., Mount, G. G., Shaffer, H. B., & Pauly, G. B. (2019). Experimental removal of introduced slider turtles offers new insight into competition with a native, threatened turtle. *PeerJ*, 7, e7444.
<https://doi.org/10.7717/peerj.7444>

Ray. (2026, March 8). *Western Painted Turtle: Complete Guide to Habitat, Diet & Life*. Private Lands Wildlife. <https://privatelandswildlife.com/western-painted-turtle/>

